

AZURE FREE TIER ACCOUNT SETUP

Complete Learning Program — Detailed Assignment Submission
ProLaunch Careers | Cloud Computing Track

This document addresses every improvement point identified in the assignment rubric for the Azure Free Tier Account Setup project. Each section provides written documentation, step-by-step procedures, and explanations designed to demonstrate full understanding of the cloud computing concepts covered in this learning program.

01

README & Repository Setup

Create a README.md file in your repository to house all written documentation

1.1 What is a README.md?

A README.md is a Markdown-formatted text file placed at the root of a GitHub repository. GitHub automatically renders it as the repository's landing page. For this assignment, the README.md serves as the central document that records every step taken during the Azure account setup process.

1.2 Step-by-Step: Creating the GitHub Repository

1. Go to <https://github.com> and sign in to your account.
2. Click the '+' icon in the top-right corner, then select 'New repository'.
3. Fill in the repository name (e.g., azure-free-tier-setup), add a short description, and choose 'Public'.
4. Check the box 'Add a README file' to auto-generate a README.md.
5. Click 'Create repository'.

TIP

Always initialize your repository with a README so you have something to edit immediately.

1.3 README.md Structure for This Assignment

Your README.md must contain the following sections as headings:

- **# Azure Free Tier Account Setup**
 - Top-level project title

- **## Overview**
 - Brief description of what Azure is and why you created an account
- **## Account Creation Steps**
 - Step-by-step written documentation with screenshots
- **## Azure Portal Navigation**
 - Description of the dashboard, service menus, and search bar
- **## Security Configuration (MFA)**
 - How to enable Multi-Factor Authentication
- **## Billing Alerts & Cost Management**
 - How to set up budget alerts and monitor spending
- **## Free Tier Limits & Summary**
 - Matrix of popular free services, quotas, and credit information
- **## Troubleshooting**
 - Common issues encountered and how they were resolved
- **## Completion Checklist**
 - All 8 required steps verified

02

Account Creation Documentation

Document the step-by-step signup process, including verification steps

2.1 Overview

Account creation is the foundation of this assignment. You must provide written instructions — not just screenshots — explaining every stage of the signup process, from navigating to the Azure website to entering payment information and verifying your identity.

2.2 Step-by-Step: Creating a Microsoft Azure Free Tier Account

6. Open your web browser and go to <https://azure.microsoft.com/en-us/free>
7. Click the green 'Start free' button on the homepage.
8. Sign in with an existing Microsoft account OR click 'Create one' to register a new account. If creating new: enter your preferred email address, create a password, and verify your email through the confirmation link sent to your inbox.
9. On the 'About you' page, fill in: First name, Last name, Country/Region, and Phone number.
10. PHONE VERIFICATION: Click 'Text me' or 'Call me'. Enter the 7-digit verification code you receive to confirm your phone number.
11. IDENTITY VERIFICATION BY CARD: Enter a valid credit or debit card. Note: Azure will charge a small temporary authorization hold (usually \$1) that is reversed within a few days. You will NOT be charged unless you upgrade from Free Tier.
12. Read and accept the Microsoft Azure Subscription Agreement and the Privacy Statement. Tick both checkboxes.
13. Click 'Sign up'. Azure will provision your account and redirect you to the Azure Portal.

IMPORTANT

Save a screenshot of each verification screen (phone OTP entry, card verification screen, and the final 'Welcome to Azure' confirmation page) and upload them to your GitHub repository.

2.3 What the Portal Home Page Shows

After account creation, your Azure Portal home page (portal.azure.com) will display:

- Your \$200 free credit balance in the top notification bar
- Subscription status: 'Free Trial' or 'Azure for Students'
- Expiration date: 30 days from account creation for the \$200 credit
- Quick-launch tiles for popular services (VMs, Storage, SQL, etc.)
- Recent resources section (empty on a new account)

03

Azure Portal Navigation

Written documentation on the dashboard, key services, and search functionality

3.1 Dashboard Overview

The Azure Portal is a unified web-based console accessible at portal.azure.com. It provides a graphical interface for managing all Azure resources. When you log in, you land on the Home page, which is different from the Dashboard. Understanding both is essential.

3.2 Key Areas of the Portal

Portal Area	Purpose & How to Use It
Home	The default landing page. Shows Azure services, recent resources, and recommended tools.
Dashboard	A customizable page made of 'tiles'. You can pin any resource, chart, or metric here.
Left Navigation Bar	Always visible. Contains: Home, Dashboard, All Services, Subscriptions, Resource Groups, and recently visited resources.
Search Bar (top)	Type any service name (e.g., 'Virtual Machines') to jump directly to it. Also searches your existing resources.
Notifications Bell	Shows alerts about deployments, billing warnings, and policy violations.
Settings (gear icon)	Change portal language, theme (dark/light), and default directory.
Cloud Shell (>_)	Opens a browser-based terminal supporting Bash or PowerShell to run Azure CLI commands.

3.3 How to Use Search Functionality

14. Click the search bar at the top of the portal (or press G then / as keyboard shortcut).
15. Type the name of a service (e.g., 'Storage accounts', 'Resource groups', 'Subscriptions').
16. The dropdown shows three categories: Services, Resources, and Documentation. Services are Azure product categories; Resources are things you have already created.
17. Click the service to open its management blade (a panel that slides in from the right).

04

Dashboard Customization

Evidence of custom dashboard creation or configuration

4.1 Why Customize the Dashboard?

A customized dashboard allows you to see the metrics and resources most relevant to your workflow at a glance. For this assignment, you must create at least one custom dashboard and document the process.

4.2 Step-by-Step: Creating a Custom Dashboard

18. In the Azure Portal, click 'Dashboard' in the left navigation bar.
19. Click '+ New dashboard' at the top of the page, then choose 'Blank dashboard'.
20. Give your dashboard a name, for example: 'My Azure Free Tier Monitor'.
21. From the Tile Gallery on the right, drag the following tiles onto your dashboard:
 - All resources — shows every resource you have created
 - Service Health — displays any Azure service outages in your region
 - Cost Management — shows your current spending vs. your budget
 - Subscription — shows your subscription status and credit remaining
22. Resize tiles by dragging their bottom-right corner. Arrange them logically.
23. Click 'Save' to save your dashboard layout.
24. Take a screenshot of your completed dashboard and upload it to your GitHub repository.

SCREENSHOT

Your screenshot must clearly show: your dashboard name, at least 4 tiles, and your real Azure account email/name visible in the top-right corner.

05

Billing & Cost Management

Documentation on cost alerts, budgets, and billing section navigation

5.1 Why Billing Awareness Matters

Even on the Free Tier, it is critical to set up billing alerts. If your free credit is exhausted or if you accidentally deploy a paid resource, Azure will begin charging your credit card. Budget alerts ensure you are notified before this happens.

5.2 Navigating the Billing Section

25. In the Azure Portal search bar, type 'Cost Management + Billing' and click the result.
26. In the left panel, explore the following sections:
 - Overview: Shows your current charges, payment due date, and subscription status.
 - Invoices: Lists all past billing statements.
 - Payment methods: Shows your registered credit/debit card.
 - Subscriptions: Shows your active subscription (Free Trial or Pay-As-You-Go).
27. Click on 'Cost analysis' in the left panel to see a breakdown of spending by service.

5.3 Step-by-Step: Setting a Budget Alert

28. In Cost Management + Billing, click 'Budgets' in the left navigation.
29. Click '+ Add' to create a new budget.
30. Set the following fields:
 - Scope: Select your subscription
 - Name: Enter a name (e.g., 'Monthly-Free-Tier-Alert')
 - Reset period: Monthly
 - Budget amount: \$5 (this triggers an alert well before your free credit runs out)
31. Under 'Set alerts', click '+ Add alert condition':
 - Alert condition: Actual
 - % of budget: 80
 - Alert recipients (email): Enter your email address
32. Click 'Create'. You will now receive an email alert if your spending reaches 80% of \$5.
33. Create a second alert at 100% of budget as a final warning.

SCREENSHOT REQUIRED

Take a screenshot of your completed budget alert setup page and include it in your README.md under the Billing section.

06

Free Tier Limits & Summary

Matrix of resource quotas, service limits, and the \$200 credit

6.1 Understanding the Azure Free Tier Offer

The Azure Free Tier has two components: (1) a \$200 credit valid for 30 days which can be used on any Azure service, and (2) a set of Always Free services that never expire. After the 30-day credit period ends, you can continue using the Always Free services at no cost.

6.2 Free Tier Resource Quotas Matrix

Service	Free Limit	Duration
Azure Virtual Machines	750 hours/month (B1s, Linux)	12 months free
Azure Blob Storage	5 GB locally redundant storage	12 months free

Azure SQL Database	250 GB storage, S0 tier	12 months free
Azure App Service	10 web, mobile, or API apps	Always free
Azure Functions	1 million requests/month	Always free
Azure Cosmos DB	1,000 RU/s throughput, 25 GB	Always free
Azure DevOps	5 users with basic access	Always free
Azure Active Directory	50,000 MAUs free tier	Always free
Azure Kubernetes Service	Cluster management free	Always free
\$200 Credit	Use on any Azure service	First 30 days only

07**Security Considerations***MFA setup, security settings, and password policies*

7.1 Why Security Configuration is Critical

Cloud accounts are high-value targets for attackers because they can spin up paid resources (cryptomining, botnets) on your credit card. Enabling Multi-Factor Authentication (MFA) and reviewing password policies are mandatory steps in any production Azure environment.

7.2 Step-by-Step: Enabling MFA on Your Azure Account

34. Sign in to the Microsoft Account security page at <https://account.microsoft.com/security>
35. Click 'Advanced security options'.
36. Under 'Two-step verification', click 'Turn on'.
37. Follow the setup wizard. You will be prompted to choose a second factor:
 - Option A (Recommended): Microsoft Authenticator app — download the app, scan the QR code shown on screen.
 - Option B: SMS text message — enter your phone number, receive a code each login.
 - Option C: Email — a verification code is sent to an alternate email.
38. Confirm your second factor by entering the code it provides.
39. Save your recovery codes in a secure location (password manager or printed and stored safely).
40. Click 'Finish'. MFA is now active on your account.

7.3 Azure Portal Security Settings

41. In the Azure Portal, search for 'Azure Active Directory' and click it.
42. Click 'Security' in the left navigation, then 'Authentication methods'.
43. Review the available authentication methods (SMS, Voice call, Authenticator app, FIDO2 keys).
44. Click 'Identity Protection' to see risk policies — review (do not change) the 'Sign-in risk policy'.

45. Navigate to 'Conditional Access' under Security to understand how access policies can be enforced.

PASSWORD POLICY

Azure enforces a minimum 8-character password with uppercase, lowercase, numbers, and special characters. Never reuse passwords between accounts. Use a password manager such as Bitwarden or 1Password.

08

Troubleshooting & Completion Checklist

Common issues, resolutions, and all 8 required steps verified

8.1 Common Issues & Resolutions

Issue	Resolution
Card declined during signup	Use a Visa or Master card debit card. Prepaid/virtual cards are often rejected. Ensure international online transactions are enabled with your bank.
Phone verification code not received	Wait 60 seconds before requesting a new code. Check your spam folder. Try 'Call me' instead of SMS.
'Account already exists' error	Use a different email address. Microsoft accounts are linked to email, not phone numbers.
Azure Free Trial already lapsed	Navigate to portal.azure.com > Subscriptions > click on your subscription > click 'Upgrade'. This converts to Pay-As-You-Go.
Cannot create Resource Group	Ensure your subscription is in 'Active' status under Subscriptions. A disabled or expired subscription blocks resource creation.
MFA prompt looping infinitely	Clear browser cache and cookies, then try signing in from a different browser or incognito window.
Budget alert email not received	Check your spam folder. Add 'microsoft.com' to your safe senders list. Verify the email address was entered correctly in the budget alert setup.

8.2 Completion Checklist — All 8 Required Steps

Verify each item before submitting your assignment. All items must be checked.

☐	Step 1 — Account Registration — Azure account created, phone and payment verified, screenshot saved.
☐	Step 2 — Portal Exploration — Dashboard, service menus, and search bar documented in README.
☐	Step 3 — Governance Setup — Primary Subscription identified, Resource Group created.

- ☐ **Step 4 — Identity Awareness** — Azure Active Directory explored, RBAC documentation reviewed and written up.
- ☐ **Step 5 — Region Selection** — Azure regions and availability zones researched; nearest region to your geographic location identified and documented.
- ☐ **Step 6 — Resource Deployment** — Basic resource (Virtual Machine or Storage Account) deployed and configured as a test.
- ☐ **Step 7 — Security Configuration** — MFA enabled, security settings reviewed, documentation complete.
- ☐ **Step 8 — README & Verification** — README.md uploaded to GitHub with 8-10 screenshots covering all areas; completion checklist included.

8.3 Support Resources

- Azure Free Tier FAQ: <https://azure.microsoft.com/en-us/free/free-account-faq/>
- Azure Documentation Hub: <https://docs.microsoft.com/en-us/azure/>
- Azure Pricing Calculator: <https://azure.microsoft.com/en-us/pricing/calculator/>
- Azure Status Page (live outages): <https://status.azure.com/>
- Microsoft Learn — Azure Fundamentals (AZ-900): <https://learn.microsoft.com/en-us/training/paths/az-900-describe-cloud-concepts/>